

1845 ▶ 1855

“... the effect was soothing, quieting and delightful beyond measure.”

Queen Victoria, after being given chloroform as an anesthetic during childbirth, 1853

1846

Painless surgery

When American surgeon William Morton removed a tumor from a patient's neck, he first used ether fumes to put the patient to sleep. During the operation, the patient felt nothing. The fumes were an anesthetic, a substance that blocks nerve signals to the brain. The discovery of anesthetics such as ether and chloroform was a major medical breakthrough.

Pipe to connect to patient's face mask



Chloroform holder

Chloroform inhaler, 1848

1847

Hand hygiene

Ignaz Semmelweis, a Hungarian physician, discovered that deaths from puerperal fever (an infectious disease that killed many women in childbirth) fell if doctors washed their hands between treating patients. Doctors were slow to follow his advice.



Ignaz Semmelweis with a patient

1847

Heat studies

German physicist Hermann von Helmholtz stated that energy cannot be created or destroyed, it can only change its form. This is the first law of thermodynamics (the branch of physics concerned with the behavior of heat).



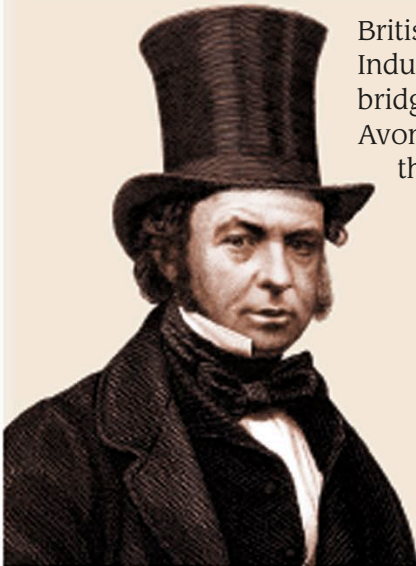
In 1848, physicist Lord Kelvin calculated the lowest possible temperature as -459.67°F (-273.15°C), which he called “absolute zero.”



1845

1806–1859

ISAMBARD KINGDOM BRUNEL



British engineer Isambard Kingdom Brunel, the greatest engineer of the Industrial Age, transformed transportation. He is best known for designing bridges, including the famous Clifton Suspension Bridge, which crosses the Avon river gorge in Bristol, UK. In 1833, he was made chief engineer of the Great Western Railway between London and Bristol.



Dangerous apprenticeship

Brunel began work at the age of 16, as his father's assistant on a project to dig a tunnel under the Thames River in London. The hazardous job was abandoned after torrential floods filled the tunnel, almost drowning Brunel.

Ocean liner

In 1845, the SS *Great Britain*, designed by Brunel, sailed from Liverpool, UK, to New York. Revolutionary in design, she was the first steam passenger liner to be built entirely of wrought iron and was driven by a massive propeller instead of paddle wheels. At the time, the *Great Britain* was the longest ship afloat. On her maiden voyage, she crossed the Atlantic in 14 days, breaking previous speed records.